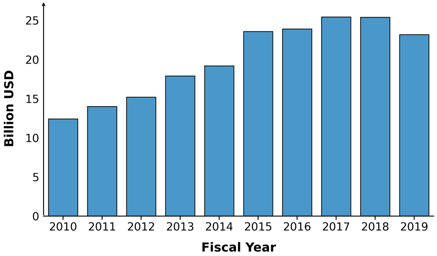
Financing, Competition and Welfare

Evidence from Small Business Loans

Maggie Isaacson and **Chinmay Lohani**

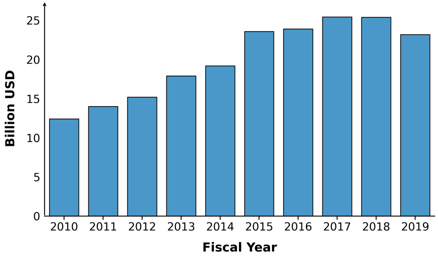
Small Businesses, Large Credit

- Assisted lending to small firms: a major industrial policy
  - Ten largest economies: \$1+ trillion (2010–2019)
  - US SBA 7(a): \$200+ billion



Small Businesses, Large Credit

- Assisted lending to small firms: a major industrial policy
  - Ten largest economies: \$1+ trillion (2010–2019)
  - US SBA 7(a): \$200+ billion



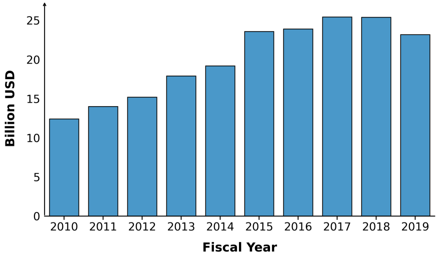
U.S. Small Business Administration

"...preserve free competitive enterprise..."

- One stated aim: **competition**
- Small firms: financing is a major challenge

Small Businesses, Large Credit

- Assisted lending to small firms: a major industrial policy
  - Ten largest economies: \$1+ trillion (2010–2019)
  - US SBA 7(a): \$200+ billion



U.S. Small Business Administration

"...preserve free competitive enterprise..."

- One stated aim: **competition**
- Small firms: financing is a major challenge
- Who competes: profitable + **financed**

If Finance Shapes Markets, Does Assistance Help?

- Loan assistance reshapes markets: helpful only if **social surplus** rises

If Finance Shapes Markets, Does Assistance Help?

- Loan assistance reshapes markets: helpful only if **social surplus** rises

Potential benefits

- allocation: fund collateral blocked firms
- competition: more firms, lower prices

If Finance Shapes Markets, Does Assistance Help?

- Loan assistance reshapes markets: helpful only if **social surplus** rises

Potential benefits

- allocation: fund collateral blocked firms
- competition: more firms, lower prices

Potential costs

- business stealing, crowd-out
- fiscal cost of the guarantee

If Finance Shapes Markets, Does Assistance Help?

- Loan assistance reshapes markets: helpful only if **social surplus** rises

Potential benefits

- allocation: fund collateral blocked firms
- competition: more firms, lower prices

Potential costs

- business stealing, crowd-out
- fiscal cost of the guarantee

- **Lender vs. social planner**:
  - lender: **private firm value**, capped by recovery
  - planner: also **consumers** and **rivals**

Research Question

How does credit assistance change market composition and welfare?

- How does small biz. credit assistance change markets via entry and exit?
- How does this impact welfare through credit allocation and competition?

Research Question

How does credit assistance change market composition and welfare?

- How does small biz. credit assistance change markets via entry and exit?
- How does this impact welfare through credit allocation and competition?

This paper:

- **Administrative data**: 7(a) awards + firm outcomes
  - evidence that 7(a) awards increase entry and reduce exit
- **Structural model**: industry dynamics with financing
  - loan market: collateral frictions $\Rightarrow$ credit constraints
  - product market: price competition $\Rightarrow$ profits + consumer surplus
  - linkage: credit $\rightarrow$ entry/exit $\rightarrow$ competition $\rightarrow$ credit
  - **Estimation:** model estimated on entry & exit data
- **Counterfactuals**: hotel industry
  - value of current 7(a) awards; alternative targeting

Preview of Findings

- 7(a) reduces exit by **22%** (5 yrs), raises entry by **17%**

Preview of Findings

- 7(a) reduces exit by **22%** (5 yrs), raises entry by **17%**
- Hotel surplus **+\$54m**, fiscal cost **\$24m**
  - allocation 3/5, competition 2/5
  - consumers \$46m vs. producers \$8m

Preview of Findings

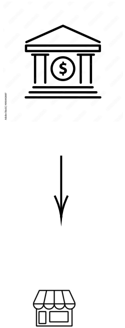
- 7(a) reduces exit by **22%** (5 yrs), raises entry by **17%**
- Hotel surplus **+$54m**, fiscal cost **$24m**
  - allocation 3/5, competition 2/5
  - consumers $46m vs. producers $8m
- Optimal allocation depends on the **objective**
  - consumer-optimal- small, concentrated markets
  - efficient- existing firms in large markets

Outline

Institutional Background: SBA 7(a)

Small Business Administration's 7(a) program

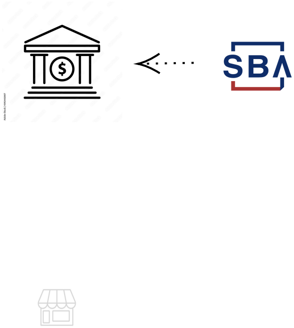
- Loans administered by private lenders



Institutional Background: SBA 7(a)

Small Business Administration's 7(a) program

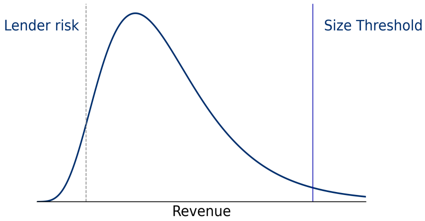
- Loans administered by private lenders
- Government guarantees a portion



Institutional Background: SBA 7(a)

Small Business Administration's 7(a) program

- Loans administered by private lenders
- Government guarantees a portion
- 7(a) Eligibility:
  - "small" within industry by rev./ emp.

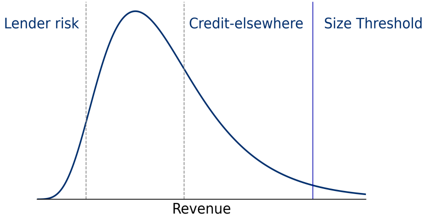


SBA places size limit, lenders reject smaller firms

Institutional Background: SBA 7(a)

Small Business Administration's 7(a) program

- Loans administered by private lenders
- Government guarantees a portion
- 7(a) Eligibility:
  - "small" within industry by rev./ emp.
  - must qualify "credit elsewhere test"; last resort



Credit elsewhere test selects smaller firms

Institutional Background: SBA 7(a)

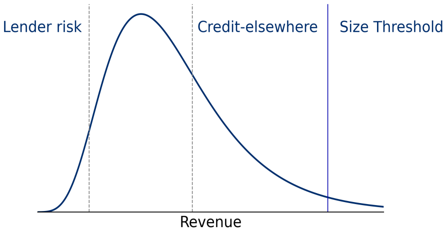
Small Business Administration's 7(a) program

- Loans administered by private lenders
- Government guarantees a portion
- 7(a) Eligibility:
  - "small" within industry by rev./ emp.
  - must qualify "credit elsewhere test"; last resort

Eligible businesses ***must not be able to obtain the desired credit on reasonable terms*** *from non-Federal, non-State, and non-local government sources.*

SBA Federal Regulation 120.101

▸ 7(a) makes loans profitable



Margins of selection

Data Sources

- **7(a) Awards**: Policy treatment.

Owner	Chinmay Lohani
Name	Cloud Abodes LLC
Address	133 S 36th St, Philadelphia
Loan Amount	\$1.2M
Int. Rate	8.75%
Year	2019
Status	Charged off
Losses	\$300k

Data Sources

- **7(a) Awards**: Policy treatment.
  - which business

Owner	Chinmay Lohani
Name	Cloudy Abodes LLC
Address	133 S 36th St, Philadelphia
Loan Amount	$1.2M
Int. Rate	8.75%
Year	2019
Status	Charged off
Losses	$300k

Data Sources

- **7(a) Awards**: Policy treatment.
  - which business
  - on what terms

Owner	Chinmay Lohani
Name	Cloud Abodes LLC
Address	133 S 36th St, Philadelphia
Loan Amount	$1.2M
Int. Rate	8.75%
Year	2019
Status	Charged off
Losses	$300k

Data Sources

- **7(a) Awards**: Policy treatment.
  - which business
  - on what terms
  - what happened to the loan

Owner	Chinmay Lohani
Name	Cloud Abodes LLC
Address	133 S 36th St, Philadelphia
Loan Amount	\$1.2M
Int. Rate	8.75%
Year	2019
Status	Charged off
Losses	\$300k

Data Sources

- **7(a) Awards**: Policy treatment.
  - which business
  - on what terms
  - what happened to the loan
- **Census Microdata**: Firm outcomes.

Owner	Chinmay Lohani
Name	Cloud Abodes LLC
Address	133 S 36th St, Philadelphia
Year	2019
Start Year	2019
End Year	2020
Revenue	$38,000
Employment	1

Data Sources

- **7(a) Awards**: Policy treatment.
  - which business
  - on what terms
  - what happened to the loan
- **Census Microdata**: Firm outcomes.
  - which business

Owner	Chinmay Lohani
Name	Cloud Abodes LLC
Address	133 S 36th St, Philadelphia
Year	2019
Start Year	2019
End Year	2021
Revenue	$38,000
Employment	1

Data Sources

- **7(a) Awards**: Policy treatment.
  - which business
  - on what terms
  - what happened to the loan
- **Census Microdata**: Firm outcomes.
  - which business
  - yearly revenue

Owner	Chinmay Lohani
Name	Cloud Abodes LLC
Address	133 S 36th St, Philadelphia
Year	2019
Start Year	2019
End Year	2021
Revenue	$38,000
Employment	1

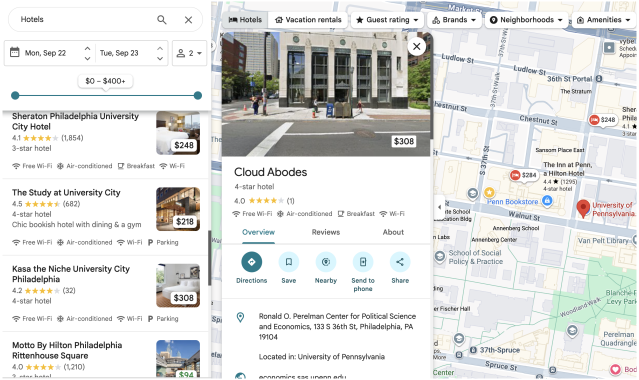
Data Sources

- **7(a) Awards**: Policy treatment.
  - which business
  - on what terms
  - what happened to the loan
- **Census Microdata**: Firm outcomes.
  - which business
  - yearly revenue
  - entry and exit

Owner	Chinmay Lohani
Name	Cloud Abodes LLC
Address	133 S 36th St, Philadelphia
Year	2019
Start Year	2019
End Year	2021
Revenue	$38,000
Employment	1

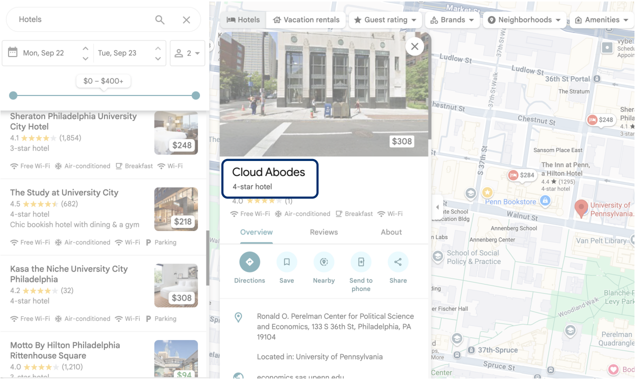
Data Sources

- **7(a) Awards**: Policy treatment.
  - which business
  - on what terms
  - what happened to the loan
- **Census Microdata**: Firm outcomes.
  - which business
  - yearly revenue
  - entry and exit
- **Hotel Prices Dataset**: Hotel details.



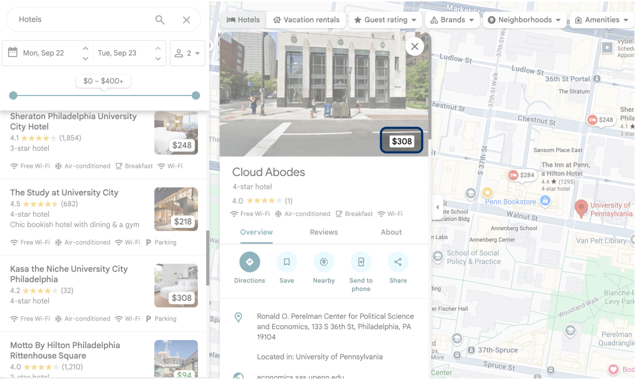
Data Sources

- **7(a) Awards**: Policy treatment.
  - which business
  - on what terms
  - what happened to the loan
- **Census Microdata**: Firm outcomes.
  - which business
  - yearly revenue
  - entry and exit
- **Hotel Prices Dataset**: Hotel details.
  - which business



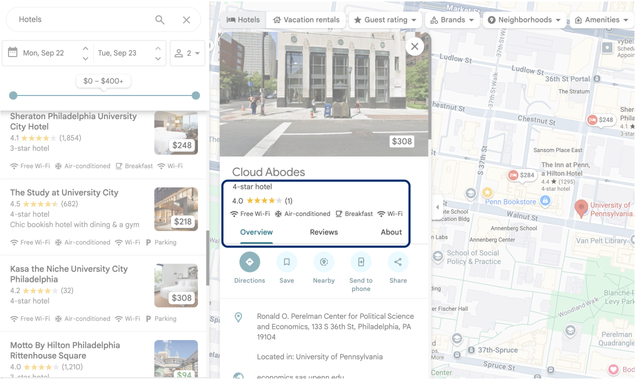
Data Sources

- **7(a) Awards**: Policy treatment.
  - which business
  - on what terms
  - what happened to the loan
- **Census Microdata**: Firm outcomes.
  - which business
  - yearly revenue
  - entry and exit
- **Hotel Prices Dataset**: Hotel details.
  - which business
  - price, features



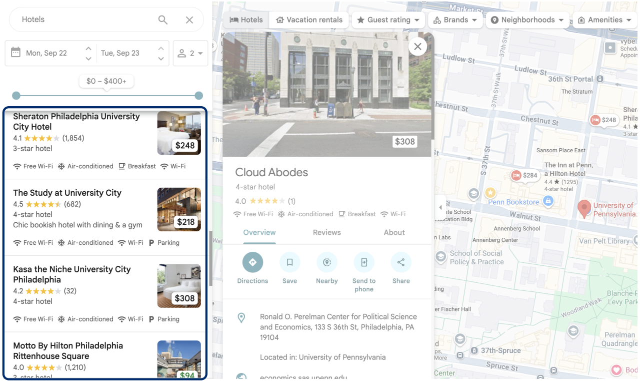
Data Sources

- **7(a) Awards**: Policy treatment.
  - which business
  - on what terms
  - what happened to the loan
- **Census Microdata**: Firm outcomes.
  - which business
  - yearly revenue
  - entry and exit
- **Hotel Prices Dataset**: Hotel details.
  - which business
  - price, features



Data Sources

- **7(a) Awards**: Policy treatment.
  - which business
  - on what terms
  - what happened to the loan
- **Census Microdata**: Firm outcomes.
  - which business
  - yearly revenue
  - entry and exit
- **Hotel Prices Dataset**: Hotel details.
  - which business
  - price, features
  - competitors, markets

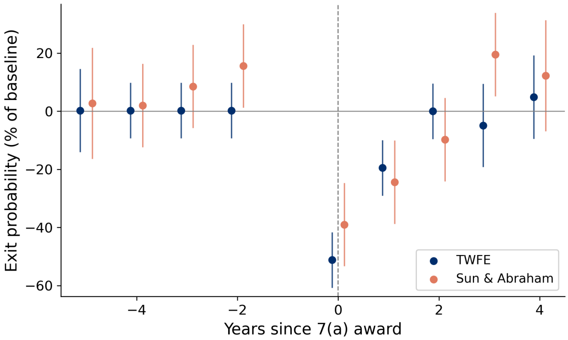


▸ Market construction ▸ Eg Philly markets

▸ US Distribution

Outline

7(a) Reduces Exit, Raises Entry



- **Exit** (matched DiD, Census): treated -22% over 5 yrs ▸ exit detail
- **Entry** (shift-share IV): 10% more 7(a) $\rightarrow$ $+1.7\%$ entry ▸ entry detail
- welfare? structural model to quantify

Outline

Primitives: Profit Stream

Uncertain Firm Profit

Creates need for debt

Lenders

Set Interest + credit limits

In incomplete markets w. default risk

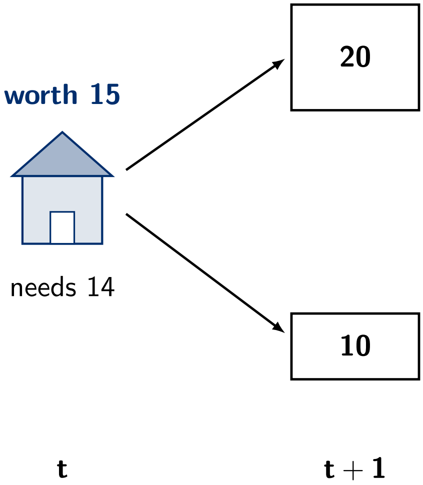
Industry dynamics

Profit

Entry and exit

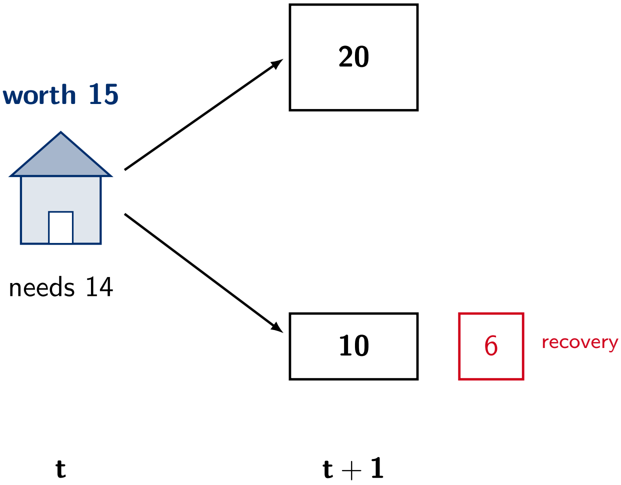
Toy Model of Credit Frictions

To **enter**, or to **keep operating**, a firm needs **14**. Next period it earns **20** or **10**.



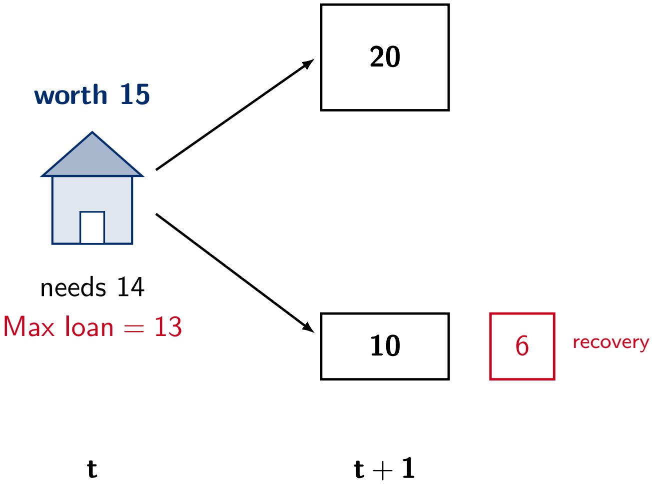
Toy Model of Credit Frictions

If it defaults, the lender recovers only **6**: its assets are depleted.



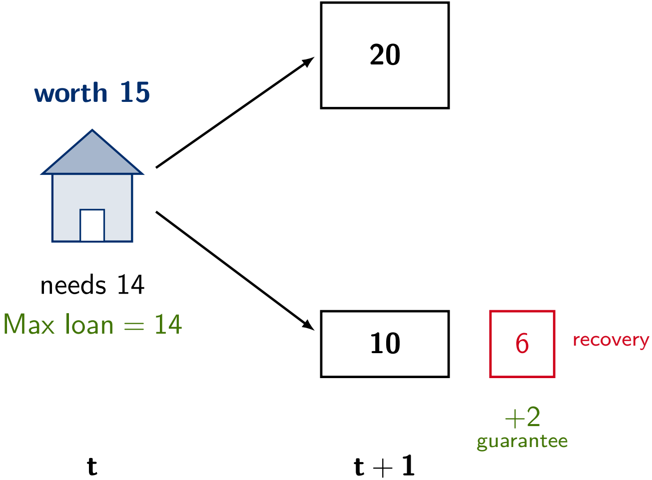
Toy Model of Credit Frictions

Most it can borrow is $(20 + 6)/2 = \mathbf{13}$, but it needs **14**. Blocked at entry; forced out as an incumbent.



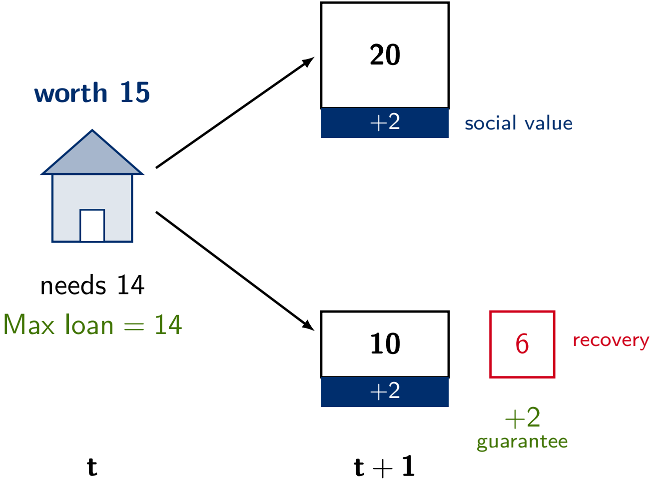
Toy Model of Credit Frictions

Guarantee tops recovery 6 → 8, ceiling rises to **14**. It enters, or it survives.



Toy Model of Credit Frictions

And each operating firm also creates social value the lender never counts.



Credit Supply: Lenders

Uncertain Firm Profit

Creates need for debt

Lenders

Set Interest + credit limits

In incomplete markets w. default risk

Industry dynamics

Profit

Entry and exit

Lender Risk and Profits

- Loans modelled as **one-period debt**: borrow L_t, repay next period

Lender Risk and Profits

- Loans modelled as **one-period debt**: borrow L_t, repay next period
- Two states for the lender:
  - repaid: $(1 + r)L_t$
  - default: ωL_t, recovery $\omega < 1$

Lender Risk and Profits

- Loans modelled as **one-period debt**: borrow L_t, repay next period
- Two states for the lender:
  - repaid: $(1+r)L_t$
  - default: ωL_t, recovery $\omega < 1$
- Competitive lenders break even $\Rightarrow$ interest rate **and** credit limit

$$\mathbb{E}\left[\pi^{\text{lender}}(r, L_t)\right] = p(r)\,\omega L_t + (1 - p(r))(1+r)L_t$$

Lender Risk and Profits

- Loans modelled as **one-period debt**: borrow L_t, repay next period
- Two states for the lender:
  - repaid: $(1+r)L_t$
  - default: ωL_t, recovery $\omega < 1$
- Competitive lenders break even $\Rightarrow$ interest rate **and** credit limit

$$\mathbb{E}\left[\pi^{\text{lender}}(r, L_t)\right] = p(r)\,\omega L_t + (1 - p(r))(1+r)L_t$$

- Guarantee raises ω: the lever the government pulls

Estimates: Volatile Profits, Tight Credit

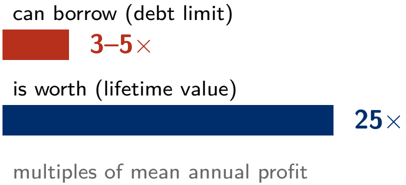
- **Demand**: nested logit
  - own-price elasticity -5.4
  - within-nest corr. $\mathbf{0.65} \Rightarrow$ business stealing

Estimates: Volatile Profits, Tight Credit

- **Demand**: nested logit
  - own-price elasticity -5.4
  - within-nest corr. **0.65** $\Rightarrow$ business stealing
- **Profits are volatile**, recovery is poor
  - demand shocks: **27%** of annual profit
  - unknown type: **64%** of profit at entry
  - recovery: **61%** of the loan

Estimates: Volatile Profits, Tight Credit

- **Demand**: nested logit
  - own-price elasticity -5.4
  - within-nest corr. **0.65** $\Rightarrow$ business stealing
- **Profits are volatile**, recovery is poor
  - demand shocks: **27%** of annual profit
  - unknown type: **64%** of profit at entry
  - recovery: **61%** of the loan
- $\Rightarrow$ firms are **highly constrained**



Industry Dynamics: Entry and Exit

Uncertain Firm Profit

Creates need for debt

Lenders

Set Interest + credit limits

In incomplete markets w. default risk

Industry dynamics

Profit

Entry and exit

Outline

Setting & Data

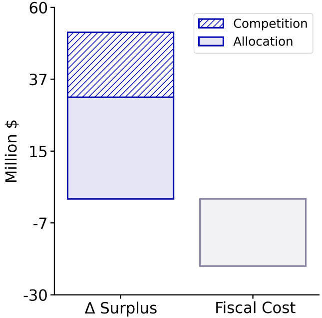
Evidence

Model

Counterfactuals

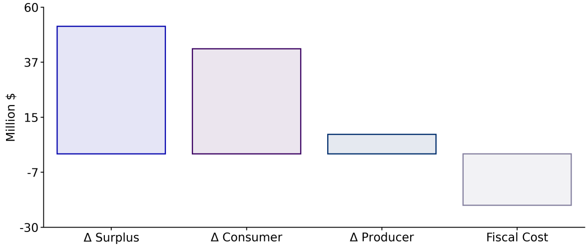
Conclusion

Value of Current 7(a)



- Raises surplus **$54m** (0.6% of revenue), cost **$24m**
  - allocation to NPV firms: $31m
  - competition: $23m

Value of Current 7(a)



- Raises surplus **$54m** (0.6% of revenue), cost **$24m**
  - allocation to NPV firms: $31m
  - competition: $23m
- consumers $46m vs. producers $8m
  - customers gain from *firm* subsidy

Smart Targeting

– Optimal policy over Revenue $\times$ HHI bins

Smart Targeting

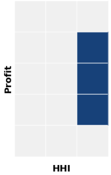
- Optimal policy over Revenue $\times$ HHI bins
- **Consumer-optimal** (maximize consumer surplus)
  - awards only in concentrated, small markets
  - multiplier **5.1**, Δ consumer welfare **+40%**

Smart Targeting

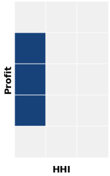
- Optimal policy over Revenue $\times$ HHI bins
- **Consumer-optimal** (maximize consumer surplus)
  - awards only in concentrated, small markets
  - multiplier **5.1**, Δ consumer welfare **+40%**
- **Efficient** (total surplus, net of producer loss)
  - large markets: support incumbents, prevent exit
  - no awards to bottom 20%: too costly
  - multiplier **5.3**

Smart Targeting

- Optimal policy over Revenue × HHI bins
- **Consumer-optimal** (maximize consumer surplus)
  - awards only in concentrated, small markets
  - multiplier **5.1**, Δ consumer welfare **+40%**
- **Efficient** (total surplus, net of producer loss)
  - large markets: support incumbents, prevent exit
  - no awards to bottom 20%: too costly
  - multiplier **5.3**
- The optimum **depends on whose surplus counts**: consumers vs. total



Consumer-optimal



Efficient

Outline

Conclusion

- Credit frictions shape market structure: who competes is who can finance

Conclusion

- Credit frictions shape market structure: who competes is who can finance
- Tractable model: financing $\rightarrow$ competition $\rightarrow$ dynamics
  - estimated on **entry and exit data**

Conclusion

- Credit frictions shape market structure: who competes is who can finance
- Tractable model: financing $\rightarrow$ competition $\rightarrow$ dynamics
  - estimated on **entry and exit data**
- 7(a) is welfare-improving: **$2.3** per $1 cost, consumers capture most
  - targeting by market structure $\rightarrow$ multiplier $\sim$**5**

Conclusion

- Credit frictions shape market structure: who competes is who can finance
- Tractable model: financing $\rightarrow$ competition $\rightarrow$ dynamics
  - estimated on **entry and exit data**
- 7(a) is welfare-improving: **\$2.3** per \$1 cost, consumers capture most
  - targeting by market structure $\rightarrow$ multiplier $\sim$**5**
- **The wedge**: private credit can't appropriate social surplus $\Rightarrow$ misallocates

Conclusion

- Credit frictions shape market structure: who competes is who can finance
- Tractable model: financing $\rightarrow$ competition $\rightarrow$ dynamics
  - estimated on **entry and exit data**
- 7(a) is welfare-improving: **$2.3** per $1 cost, consumers capture most
  - targeting by market structure $\rightarrow$ multiplier $\sim$**5**
- **The wedge**: private credit can't appropriate social surplus $\Rightarrow$ misallocates
- Financial policy: a lever on market structure, promising for welfare

"Could the banks revive Britain's economy?"
Financial Times, Nov 2025

"America's two-track economy: the small business credit crunch"
Financial Times, Dec 2020

Thanks!

chinmaylohani.com

7(a) Increases Entry

- Markets w. more assistance $\Rightarrow$ more entry?

  $$y_{mt} = \beta^{entry} x_{mt} + \Gamma' W_{mt} + \gamma_t + \mu_m + \varepsilon_{mt}$$

  - y_{mt}: log(#entrants) in market m, year t
  - x_{mt}: log(7(a) dollars)
  - W_{mt} lagged avg. emp., wage

7(a) Increases Entry

- Markets w. more assistance $\Rightarrow$ more entry?

$$y_{mt} = \beta^{entry} x_{mt} + \Gamma' W_{mt} + \gamma_t + \mu_m + \varepsilon_{mt}$$

  - y_{mt}: log(#entrants) in market m, year t
  - x_{mt}: log(7(a) dollars)
  - W_{mt} lagged avg. emp., wage

- IV: shift-share (Greenstone et al. (2020))
  - **shifts**: (residualized) national lending by bank
  - shares: 2009 share of city m in a bank's 7(a)

IV Details   Robustness Checks

7(a) Increases Entry

- Markets w. more assistance $\Rightarrow$ more entry?

$$y_{mt} = \beta^{entry} x_{mt} + \Gamma' W_{mt} + \gamma_t + \mu_m + \varepsilon_{mt}$$

  - y_{mt}: log(#entrants) in market m, year t
  - x_{mt}: log(7(a) dollars)
  - W_{mt} lagged avg. emp., wage

- IV: shift-share (Greenstone et al. (2020))
  - **shifts**: (residualized) national lending by bank
  - shares: 2009 share of city m in a bank's 7(a)

IV Details   Robustness Checks

	(1)	(2)	(3)	(4)
log(SBA)	0.585	0.454	0.226	0.176
	(0.013)	(0.041)	(0.076)	(0.073)
Fixed effects	Yes	Yes	Yes	Yes
IV		Yes	Yes	Yes
Quasidiff IV			Yes	Yes
City trends				Yes
N	7,200	7,200	7,200	7,200

7(a) Reduces Exit

- 7(a) award $\Rightarrow$ less exit?
- Matching + DiD
  - control firms: similar 7(a) propensity
  - DiD with 7(a) award as event

7(a) Reduces Exit

- 7(a) award $\Rightarrow$ less exit?
- Matching + DiD
  - control firms: similar 7(a) propensity
  - DiD with 7(a) award as event

$$Y_{it} = \alpha_i + \gamma_t + \Phi' X_{it} + \sum_{k \in \mathcal{K} \setminus \{-1\}} \beta_k D_i \tau_k + \varepsilon_{it}$$

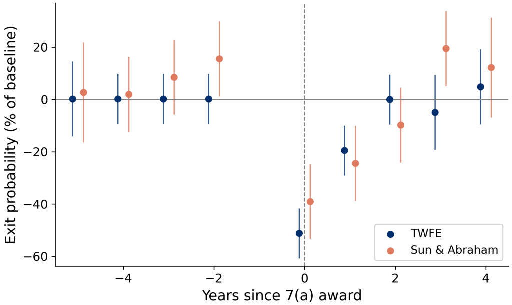
  - Y_{it}: exit indicator
  - D_i: ever-treated; τ_k: event-time dummies
  - X_{it}: city/year/industry/age-bin FE + controls (emp., rev.)

7(a) Reduces Exit

- 7(a) award $\Rightarrow$ less exit?
- Matching + DiD
  - control firms: similar 7(a) propensity
  - DiD with 7(a) award as event

$$Y_{it} = \alpha_i + \gamma_t + \Phi' X_{it} + \sum_{k \in \mathcal{K} \setminus \{-1\}} \beta_k D_i \tau_k + \varepsilon_{it}$$

  - Y_{it}: exit indicator
  - D_i: ever-treated; τ_k: event-time dummies
  - X_{it}: city/year/industry/age-bin FE + controls (emp., rev.)



	Coefficient	Exit Rate
Treatment Effect	-0.009	
	(0.001)	
Control Group (Post)		0.041
Treated Group (Post)		0.032
N	282,000	

Matching details | Other parallel trends | Sun & Abraham | Hotel Exit | Age Heterogeneity | Robustness to size

Identification details (shares $\times$ shifts)

Instrument

$$Z_{mt} = \sum_b s_{mb}^{\text{pre}}\, \tilde{B}_{bt}, \qquad \tilde{B}_{bt} \equiv \text{ idiosyncratic shift in bank } b\text{'s lending}$$

Relevance

- x_{mt} (local 7(a) lending) and Z_{mt} (shares $\times$ shifts) are correlated.
- Example: 2015 lending by TD in Philadelphia is correlated with its 2009 city share.

Exclusion

- ε_{mt} (local unobservables) is uncorrelated with Z_{mt}.
- Idiosyncratic national shifts in TD have no relation with Philadelphia-level shocks, after controls.

Back

Entry Results: Robustness

Non-Parametric Trends

	(1)	(2)
$\log(\mathrm{SBA})$	0.176	0.283
	(0.073)	(0.047)
Linear trend	Yes	
NP trend		Yes

Results strengthen with flexible trends

Placebo: Lead Outcome

	y_t	y_{t+1}
$\log(\mathrm{SBA})$	0.176	0.081
	(0.073)	(0.191)
Same spec.		

No effect on future entry

2015 Ceiling Increase

	Full	2014–15
$\log(\mathrm{SBA})$	0.176	0.151
	(0.073)	(0.142)
Ceiling: \$19B $\rightarrow$ \$25B		

Similar estimate; less power

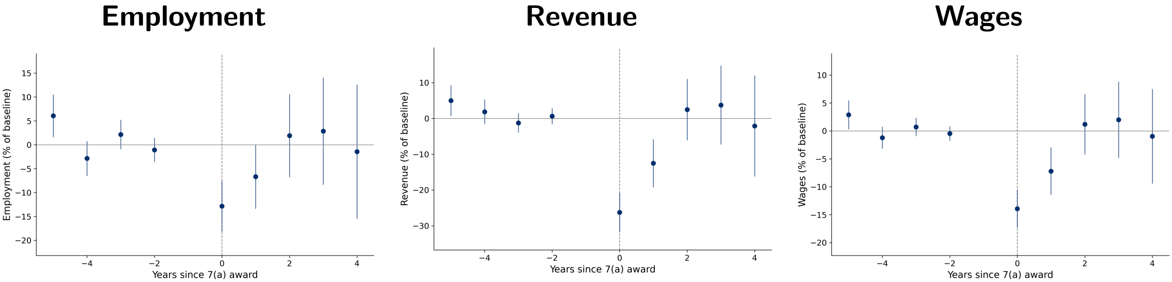
Back

Matching details

- Controls drawn from unaffected markets- SUTVA
  - no 7(a) award within industry $\times$ city between 2010-2019
- Matching within industry $\times$ age $\times$ year bins
- Propensity scores for 7(a) award based on
  - baseline revenue, employment,
  - both levels and growth

Back

Parallel Trends: Other Outcomes

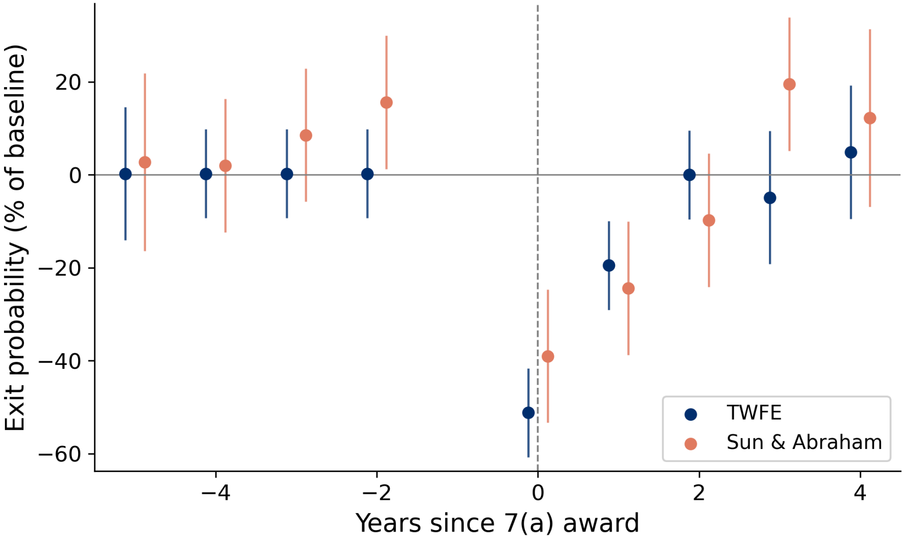


Matched on *Matched on* *Not matched on*

- No differential pre-trends in matched variables (employment, revenue)
- Wages also parallel despite not being matched on

Back

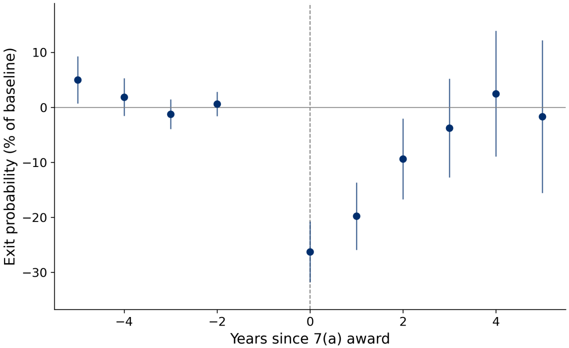
Sun & Abraham



Back

Exit Results: Hotels

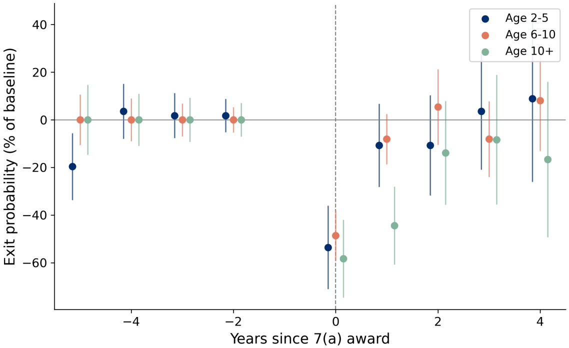
- Same specification, hotel industry only
- Stronger effects
- More persistent: significant through year 3



Back

Exit Results: Heterogeneity by Firm Age

- Effects present across all age groups
- Younger firms (age 2–5): slightly stronger initial effect
- Older firms (10+): more persistent



Back

Exit Results: Robustness to Firm Size

Concern: Selection on unobservables

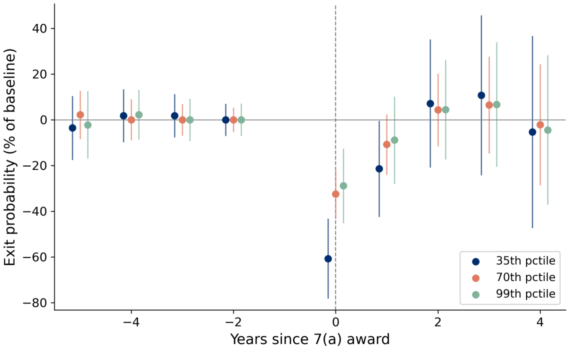
- SBA's "credit elsewhere" test ⇒ treated firms have worse collateral than average
- But some controls may be *even worse*—too risky even with guarantees

Test: If bad controls drive results...

- Smallest firms are riskiest (highest exit)
- Effects should concentrate there

Finding: Effects similar across size distribution

- Results not driven by smallest firms



By revenue percentile

Back